

EX-8

1)

```
mysql> CREATE TABLE Department (  
  ->     DeptID INT PRIMARY KEY,  
  ->     DeptName VARCHAR(30)  
  -> );  
Query OK, 0 rows affected (0.16 sec)  
  
mysql>  
mysql> CREATE TABLE Faculty (  
  ->     FID INT PRIMARY KEY,  
  ->     FName VARCHAR(30),  
  ->     DeptID INT,  
  ->     FOREIGN KEY (DeptID) REFERENCES Department(DeptID)  
  -> );  
Query OK, 0 rows affected (0.20 sec)  
  
mysql> INSERT INTO Department VALUES  
  -> (1, 'CSE'),  
  -> (2, 'ECE'),  
  -> (3, 'IT'),  
  -> (4, 'MECH');  
Query OK, 4 rows affected (0.08 sec)  
Records: 4  Duplicates: 0  Warnings: 0  
  
mysql>  
mysql> INSERT INTO Faculty VALUES  
  -> (101, 'Ramesh', 1),  
  -> (102, 'Kamal', 2),  
  -> (103, 'Priya', 3),  
  -> (104, 'Arun', 2);  
Query OK, 4 rows affected (0.00 sec)  
Records: 4  Duplicates: 0  Warnings: 0
```

INNER JOIN

```
mysql> SELECT F.FName, D.DeptName  
  -> FROM Faculty F  
  -> INNER JOIN Department D  
  -> ON F.DeptID = D.DeptID;  
  
+-----+-----+  
| FName | DeptName |  
+-----+-----+  
| Ramesh | CSE      |  
| Kamal  | ECE      |  
| Priya  | IT       |  
| Arun   | ECE      |  
+-----+-----+  
4 rows in set (0.01 sec)
```

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EQUI JOIN

```
mysql> SELECT F.FName, D.DeptName
-> FROM Faculty F, Department D
-> WHERE F.DeptID = D.DeptID;
```

FName	DeptName
Ramesh	CSE
Kamal	ECE
Priya	IT
Arun	ECE

4 rows in set (0.02 sec)

LEFT JOIN

```
mysql> SELECT F.FName, D.DeptName
-> FROM Faculty F
-> LEFT JOIN Department D
-> ON F.DeptID = D.DeptID;
```

FName	DeptName
Ramesh	CSE
Kamal	ECE
Priya	IT
Arun	ECE

4 rows in set (0.04 sec)

RIGHT JOIN

```
mysql> SELECT F.FName, D.DeptName
-> FROM Faculty F
-> RIGHT JOIN Department D
-> ON F.DeptID = D.DeptID;
```

FName	DeptName
Ramesh	CSE
Kamal	ECE
Arun	ECE
Priya	IT
NULL	MECH

5 rows in set (0.03 sec)

```
mysql>
```

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2)

```
mysql> CREATE TABLE Student (
  ->     SID INT PRIMARY KEY,
  ->     SName VARCHAR(30)
  -> );
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> CREATE TABLE Course (
  ->     CID VARCHAR(10) PRIMARY KEY,
  ->     CourseName VARCHAR(50)
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> CREATE TABLE Score (
  ->     SID INT,
  ->     CID VARCHAR(10),
  ->     Score INT,
  ->     FOREIGN KEY (SID) REFERENCES Student(SID),
  ->     FOREIGN KEY (CID) REFERENCES Course(CID)
  -> );
Query OK, 0 rows affected (0.07 sec)

mysql> INSERT INTO Student VALUES
  -> (1, 'Naveen'),
  -> (2, 'Praveen'),
  -> (3, 'Divya'),
  -> (4, 'Arun');
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql>
mysql> INSERT INTO Course VALUES
  -> ('C001', 'DBMS'),
  -> ('C002', 'Operating Systems'),
  -> ('C003', 'Computer Networks');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql>
mysql> INSERT INTO Score VALUES
  -> (1, 'C001', 85),
  -> (1, 'C002', 78),
  -> (2, 'C001', 90),
  -> (2, 'C002', 88),
  -> (3, 'C001', NULL),
  -> (3, 'C002', NULL),
  -> (4, 'C003', 75);
Query OK, 7 rows affected (0.01 sec)
Records: 7 Duplicates: 0 Warnings: 0
```

LEFT JOIN

```
mysql> SELECT S.SName, SC.CID
  -> FROM Student S
  -> LEFT JOIN Score SC
  -> ON S.SID = SC.SID
  -> WHERE SC.Score IS NULL;

+-----+-----+
| SName | CID  |
+-----+-----+
| Divya | C002 |
| Divya | C001 |
+-----+-----+
2 rows in set (0.01 sec)
```

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3)

CROSS JOIN:

```
mysql> SELECT t1.id, t2.id
-> FROM t1
-> CROSS JOIN t2;
+-----+-----+
| id    | id    |
+-----+-----+
| 3     | 10    |
| 2     | 10    |
| 1     | 10    |
| 3     | 20    |
| 2     | 20    |
| 1     | 20    |
| 3     | 30    |
| 2     | 30    |
| 1     | 30    |
+-----+-----+
9 rows in set (0.00 sec)
```

LEFT JOIN:

```
mysql> SELECT t1.id, t2.id
-> FROM t1
-> LEFT JOIN t2
-> ON t1.pattern = t2.pattern
-> ORDER BY t1.id;
+-----+-----+
| id    | id    |
+-----+-----+
| 1     | 10    |
| 2     | 20    |
| 3     | 30    |
+-----+-----+
3 rows in set (0.00 sec)
```

RIGHT JOIN:

```
mysql> SELECT t1.id, t2.id
-> FROM t1
-> RIGHT JOIN t2
-> ON t1.pattern = t2.pattern
-> ORDER BY t2.id;
+-----+-----+
| id    | id    |
+-----+-----+
| 1     | 10    |
| 2     | 20    |
| 3     | 30    |
+-----+-----+
3 rows in set (0.00 sec)
```